

CSCI 131 — Introduction to Programming in Python

Programming Assignment 2

Remainders, Powers, and Strings

This assignment covers integer division, modulus, reassignment, exponentiation, and string operators.

Problem 1 — Making Change

Write a program `change.py` which asks the user for a whole number of cents and prints how many quarters, dimes, nickels, and pennies make up that amount, using the largest coins first. Use the prompt 'Enter an amount in cents: '.

A sample run:

```
Enter an amount in cents: 187
Quarters: 7
Dimes: 1
Nickels: 0
Pennies: 2
```

Some sample solutions:

Cents	Quarters	Dimes	Nickels	Pennies
187	7	1	0	2
99	3	2	0	4
41	1	1	1	1
100	4	0	0	0
6	0	0	1	1

Problem 2 — Growth

Write a program `growth.py` which asks the user for a starting amount, a growth rate per period given as a decimal, and a number of periods. It then prints the final amount, which is

$$\text{final} = \text{start} \times (1 + \text{rate})^{\text{periods}}.$$

You should use the input prompts 'Starting amount: ', 'Growth rate per period: ', and 'Number of periods: ', in that order.

A sample run:

```
Starting amount: 200
Growth rate per period: 0.5
Number of periods: 3
Final amount: 675.0
```

Some sample solutions:

Start	Rate	Periods	Output
200	0.5	3	Final amount: 675.0
100	1.0	4	Final amount: 1600.0
1000	0.5	2	Final amount: 2250.0
50	0.2	2	Final amount: 72.0

Problem 3 — Receipt

Write a program `receipt.py` which asks the user for an item name, a quantity, and a unit price. It then prints a line of thirty hyphens, a label line naming the item and how many were bought, and the total cost. Use the prompts 'Item name: ', 'Quantity: ', and 'Unit price: ', in that order.

A sample run:

```
Item name: Notebook
Quantity: 3
Unit price: 2.50
-----
Notebook x3
Total: 7.5
```

Some sample solutions:

Item	Quantity	Price	Label line	Total line
Notebook	3	2.50	Notebook x3	Total: 7.5
Pens	12	0.75	Pens x12	Total: 9.0
Backpack	1	45.00	Backpack x1	Total: 45.0
Folders	5	1.20	Folders x5	Total: 6.0

The label line is the item name, a space, the letter `x`, and the quantity. Build it with string concatenation, and build the hyphen line with the string repetition operator. Do not type thirty hyphens by hand.

Some hints

Problem 1 counts coins, and a coin cannot be fractional, so convert with `int`. Problem 2 works with amounts and a rate, which certainly can be fractional, so convert those with `float`, but the number of periods is a count, so it is an `int`. Problem 3 has both: the quantity is an `int`, the unit price is a `float`.

Submission

Submit all three source files on Canvas by the end of the period. Put your name in a comment at the top of each file. If a classmate helped you, put their name in the comment as well.

If you are not finished when the period ends, submit what you have — a partial program earns partial credit, an empty one earns none. You may then complete it on your own and resubmit by Wednesday, September 16 at 11:59 pm for up to 50% of the remaining credit.